Part 5 — What the field has actually made: five product classes and the constraints each one imposes (weeks 53 to 57)

Phase 4 asks you to build a mental library of what has been achieved. The risk in these weeks is that the library becomes a list of anecdotes — artemisinin, butanediol, LanzaTech — which is what most candidates arrive with and what no interviewer is impressed by. So these chapters are organised the other way round: each product class is treated as a set of constraints, and every case study is there because it illustrates one of them. Each chapter ends with a constraint sheet, which is the form the host chapters of part 6 will be written against.

The single most useful habit for this phase is the one from part 2 chapter 18: convert every reported achievement into titer, yield as a fraction of theoretical, and rate, and then ask what price the product must clear. Done consistently, that reduces fifty papers to a table you can reason about and makes the pattern visible — which is that the technical difficulty of a pathway and the commercial difficulty of a product are almost uncorrelated. Twenty-one heterologous genes in yeast is a spectacular achievement at micrograms per litre; a four-gene commodity route is a decade of process work at a hundred grams per litre, and only one of them was ever a business.

Everything quantitative here is computed from stoichiometry, with the balance written out so you can check it. The reported titers are from the papers the tracker names, and are marked as reported rather than recomputed.

Ch.	Week	Chapter
53	53	Pharmaceuticals, terpenoids and natural products: the expensive-molecule regime
54	54	Commodity chemicals and amino acids: the cheap-molecule regime
55	55	Biofuels, C1 and gas fermentation, and food ingredients
56	56	PHA, from pathway to polymer: the case study worked all the way through
57	57	CO_2, hydrogen and waste carbon: when electricity is the feedstock

Pharmaceuticals, terpenoids and natural products: the expensive-molecule regime

When the product sells for thousands a kilogram, yield stops mattering and everything else starts: whether the pathway works at all, whether the enzymes are specific, and whether the molecule is worth making biologically rather than chemically.

The regime, before the case studies

Part 2 chapter 18 established that for a commodity, yield binds. Invert the price and the constraints invert with it. At $1,000/kg, glucose at a yield of 0.05 g/g costs $8 per kilogram of product, which is a rounding error, so nobody optimises yield. What binds instead is whether the pathway can be assembled at all — twenty enzymes from four organisms, several of them membrane-bound plant cytochrome P450s — whether those enzymes are specific enough not to make three side products for every product, and whether the resulting titer clears the floor set by the cost of processing broth. That floor is real but low: at 1 g/L a pharmaceutical is a viable process, where a commodity at 1 g/L is a demonstration.

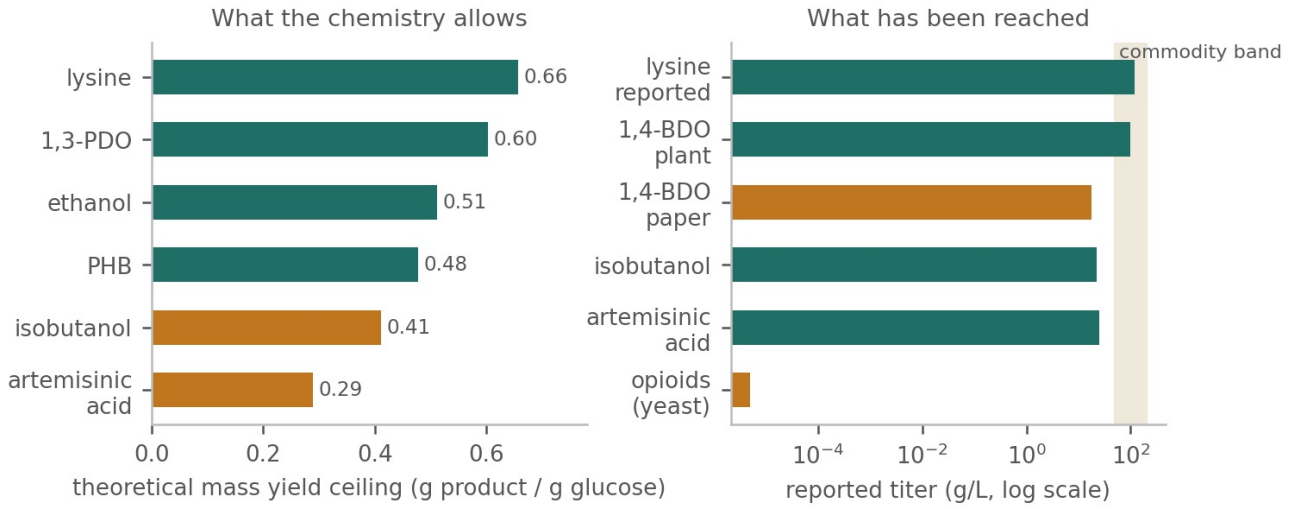


Figure 53.1 Left: theoretical mass-yield ceilings computed from the stoichiometry in this part, with the two natural-product-style targets in orange. Right: reported titers across the case studies of weeks 53 to 55, on a log scale, against the band a commodity process must reach. The spread is nine orders of magnitude, and it does not track pathway sophistication. Ceilings computed for this module; titers are as reported in the cited papers.

Artemisinic acid, and what 25 g/L actually required

The semi-synthetic artemisinin project is the field's set piece, and the reason to know it in detail is that it contains every lesson in the regime. The target: artemisinin, the frontline antimalarial, previously extracted from *Artemisia annua* with a supply chain that swung in price by factors of several. The route: engineer *S. cerevisiae* to overproduce farnesyl pyrophosphate through the mevalonate pathway, add amorphadiene synthase to cyclise it, add the plant cytochrome P450 CYP71AV1 with its reductase to oxidise the product three times to artemisinic acid, and then convert that chemically to artemisinin. Paddon and colleagues reported 25 g/L of artemisinic acid.

Work out what that number costs. Each five-carbon isoprene unit requires three acetyl-CoA, three ATP and two NADPH, and loses one carbon as CO_2 at the mevalonate decarboxylation step. A C15 sesquiterpene therefore takes nine acetyl-CoA, nine ATP and six NADPH before any oxidation. In yeast, cytosolic acetyl-CoA is made by the acetyl-CoA synthetase route, which cleaves ATP to AMP — two equivalents each, as part 1 chapter 4 insisted — so add eighteen more ATP equivalents. Then three P450 oxidations consume three more NADPH and three O_2. Totals per molecule of artemisinic acid:

```
9 acetyl-CoA  ·  27 ATP equivalents  ·  9 NADPH  ·  3 O₂  ·  3 CO₂ lost to the pathway itself
```

Nine acetyl-CoA means 4.5 glucose, so the theoretical mass yield is 234.3/(4.5 × 180.2) = **0.289 g/g** and the carbon yield is 56%. Reaching 25 g/L at the ceiling would consume 86 g/L of glucose, and the real process consumed considerably more, because 27 ATP equivalents per molecule is an enormous energy demand that has to be met by respiring carbon that then cannot become product. This is the quantitative content of the sentence everyone repeats about terpenoids being acetyl-CoA and NADPH sinks.

Two further lessons from the same paper. The P450 was the hard part — plant P450s in yeast need their reductase partner, haem, and a membrane environment, which is precisely the failure mode part 1 chapter 2 listed as the second hypothesis for an inactive band on a gel. And the process is *semi*-synthetic for a good reason: the final step from artemisinic acid to artemisinin is a photochemical oxidation that chemistry does cleanly and cheaply, and biology would do badly. The engineering judgement — make the complicated carbon skeleton biologically and the simple transformation chemically — is one of the most transferable ideas in the field, and it is the answer to give when asked where biology should stop.

Opioids in yeast, and what a complete pathway is worth

Galanie and colleagues reported complete biosynthesis of opioids in yeast from sugar, assembling 21 to 23 heterologous genes from plants, bacteria and rat. As an engineering feat it is extraordinary: the pathway spans two compartments, requires several P450s, an enzyme with a stereochemical inversion, and a reticuline epimerase that had only recently been identified. The reported titers were a few micrograms per litre.

Put that in the arithmetic of this module. At 5 µg/L you would need roughly 200,000 m³ of broth to make one kilogram of product — not a process, by five or six orders of magnitude. And yet the paper mattered, for two reasons worth separating. Scientifically, it established that the full pathway functions in a microbe, which converts the remaining work from discovery into optimisation. Politically, it triggered the field's standard dual-use discussion, because a yeast strain that makes a controlled substance from sugar is a different kind of object from a yeast strain that makes a terpene.

You should have a considered position on that, because it is an interview question and both available clichés are poor. Alarm ignores that the titers were a millionfold from usable, that poppy cultivation is not the limiting factor in opioid supply, and that the published strains are traceable, licensed and surveilled. Dismissal ignores that the gap between a microgram and a gram per litre has historically been closed in a few years, that strain distribution is harder to control than plant cultivation, and that governance regimes work better when designed before capability arrives than after. A defensible answer names both and lands somewhere specific: that the research should continue in licensed hands, that strain registries and sequence screening are proportionate, and that the field's own early engagement with the question is the reason the discussion has been calm.

The isoprenoid platform, and why the same pathway keeps appearing

Terpenoids are the largest class of natural products, and all of them are built from the same two five-carbon units — isopentenyl pyrophosphate and its isomer — by the same chemistry: condensation to C10, C15, C20 and higher prenyl chains, cyclisation by a terpene synthase, then decoration by P450s, dehydrogenases and transferases. This is why one engineered chassis serves many products: get a strain with a high flux to farnesyl pyrophosphate and you have a platform, and the product-specific work is one synthase plus a handful of tailoring enzymes.

Five commercial classes sit on that platform: flavours and fragrances such as nootkatone and valencene; pharmaceutical precursors like artemisinic acid and taxadiene; nutraceuticals and pigments including carotenoids such as astaxanthin and β-carotene; sterols and squalene for cosmetics and vaccine adjuvants; and cannabinoids, which are polyketide–terpenoid hybrids. The enzyme families recur across all of them — a prenyltransferase, a terpene synthase, a P450 with a reductase, a glycosyltransferase — which is why experience in one transfers.

Two routes exist to the five-carbon units and the choice has consequences from part 2 chapter 19: the mevalonate pathway, from acetyl-CoA, costs ATP and NADPH and loses a carbon; the methylerythritol phosphate pathway, from pyruvate and glyceraldehyde-3-phosphate, is more carbon-efficient but is tightly regulated in its native hosts and harder to de-repress. Yeast has the first, *E. coli* the second, and engineered strains frequently carry a heterologous copy of the other.

Technical success against market success

Semi-synthetic artemisinin worked and did not take the market. The reason is instructive and is not a failure of the engineering: plant-derived artemisinin remained competitive, partly because the agricultural supply responded to the same high prices that motivated the project, and partly because a fermentation route's cost floor includes capital that farmers do not pay. A programme that takes six years to deliver is competing against the price its own existence helped to stabilise.

The interview answer that uses this well avoids both cynicism and naivety. Something like: the project achieved what it set out to do technically and demonstrated that a complex plant natural product could be made at industrial scale, which changed what the field attempts; commercially it ran into the fact that an agricultural incumbent with a volatile price is a moving target, and the lesson the field took was to choose targets where the incumbent cannot respond — molecules that are hard to extract, hard to synthesise, or not available at all. That is also why the field's attention moved toward novel molecules and food ingredients rather than drop-in replacements for crops.

Constraint sheet — pharmaceuticals and natural products

Binding metric	Total output and purity. Yield on sugar is nearly irrelevant; titer matters only through processing cost.
Yield ceiling	Low by commodity standards — 0.29 g/g for a C15 terpenoid acid — and nobody optimises it.
Cofactor demand	Severe: 27 ATP equivalents and 9 NADPH per molecule for artemisinic acid. Acetyl-CoA and NADPH supply dominate strain design.
Pathway difficulty	High: 10–25 heterologous genes, plant P450s needing reductase, haem and membrane, compartment-spanning routes.
Host requirements	Eukaryote usually: endoplasmic reticulum for P450s, compartments, post-translational capability. Yeasts dominate.
Toxicity and removal	Hydrophobic and often membrane-toxic — part 3 chapter 45's regime; an overlay is standard, not optional.
Where it fails	Enzyme promiscuity producing side products, low-flux bottlenecks at tailoring steps, and an agricultural incumbent that can cut its price.

Three things people get wrong

- ‘Twenty-three genes in yeast is the hard part.’ Assembly is now routine; flux through the tailoring steps and enzyme specificity are the hard parts.
- ‘Artemisinin failed.’ It succeeded technically and lost commercially to an agricultural incumbent whose price moved — a different and more useful lesson.
- ‘Yield doesn't matter for pharmaceuticals, so stoichiometry doesn't either.’ The ATP and NADPH balance still decides whether the strain can run at all.

Self-check — answer on paper first

1. Compute the acetyl-CoA, ATP and NADPH demand of a C20 diterpene made through the mevalonate pathway in yeast, and say which supply you would engineer first.

Four C5 units: 12 acetyl-CoA, 12 ATP for the mevalonate steps, 8 NADPH, with 4 CO_2 lost at decarboxylation; plus 24 ATP equivalents if the acetyl-CoA comes through acetyl-CoA synthetase, giving 36 ATP equivalents in total. Acetyl-CoA first, because it is both the carbon and, through the synthetase route, most of the energy demand — which is why the standard interventions are a PDH-bypass replacement, routing through a cytosolic acetylating acetaldehyde dehydrogenase or an ATP-cheaper acetyl-CoA source, before touching the terpene synthase.

2. Why is a plant P450 the step most likely to fail, and what are the three fixes in order of cost?

Because it needs a membrane to insert into, haem, and a cytochrome P450 reductase partner at a matched ratio, and because its native environment is a plant endoplasmic reticulum at 25 °C. Cheapest first: co-express the reductase and supplement or upregulate haem synthesis. Then engineer the N-terminal membrane anchor, or swap to a homologue from a better-behaved organism. Most expensive: change host or compartment, or replace the P450 with a different enzyme chemistry altogether.

3. Give the argument for stopping the biology at artemisinic acid rather than going to artemisinin.

Because the remaining transformation is a photochemical oxidation and rearrangement that chemistry performs cleanly, cheaply and at scale, whereas biology would need several enzymes for a reaction with an unstable intermediate and would be slower and less selective. The general principle is to divide the synthesis where each discipline has the advantage: biology builds the chiral, polycyclic carbon skeleton from a cheap feedstock, chemistry performs the simple high-energy transformations. Insisting on a fully biological route is an aesthetic preference, not an engineering one.

What to open a book for. Three papers and no book: Paddon et al., Nature 496:528 (2013) for the build, Paddon & Keasling, Nat. Rev. Microbiol. 12:355 (2014) for the retrospective, and Galanie et al., Science 349:1095 (2015) for the opioid pathway and the governance discussion published around it. Log all three with the paper log's protocol; they recur in every interview in this field.

Commodity chemicals and amino acids: the cheap-molecule regime

Here yield is everything, the incumbent is a petrochemical plant with its capital already written down, and the strain is the easy part of a ten-year programme.

1,3-Propanediol: the archetype of a combined pathway

The DuPont bio-PDO process is the cleanest example of building a route that no organism has. The logic: *S. cerevisiae* has the genes to turn dihydroxyacetone phosphate into glycerol; *Klebsiella* has the genes to turn glycerol into 1,3-propanediol through a B_{12}-dependent dehydratase; neither organism does both. Put both halves into *E. coli* and you have glucose to PDO in one strain, which goes into Sorona polymer and won the 2003 Presidential Green Chemistry Challenge award for greener reaction conditions.

The interesting engineering is not the pathway, it is the energy. Compute the ceiling yourself, and the shape of the problem appears immediately. Glucose to two dihydroxyacetone phosphate costs 2 ATP; each DHAP to glycerol consumes one NADH; each glycerol to PDO consumes another. So two PDO per glucose requires **four NADH, and that route produces none**, because the NADH-generating step of glycolysis is downstream of the branch point. The redox balance must be closed by oxidising part of the feedstock: with x the fraction of glucose routed to product and full oxidation of the remainder yielding about 10 NADH per glucose, $4x = 10(1 - x)$ gives $x = 0.71$, so 1.43 mol of PDO per mole of glucose and a mass yield of **0.60 g/g**.

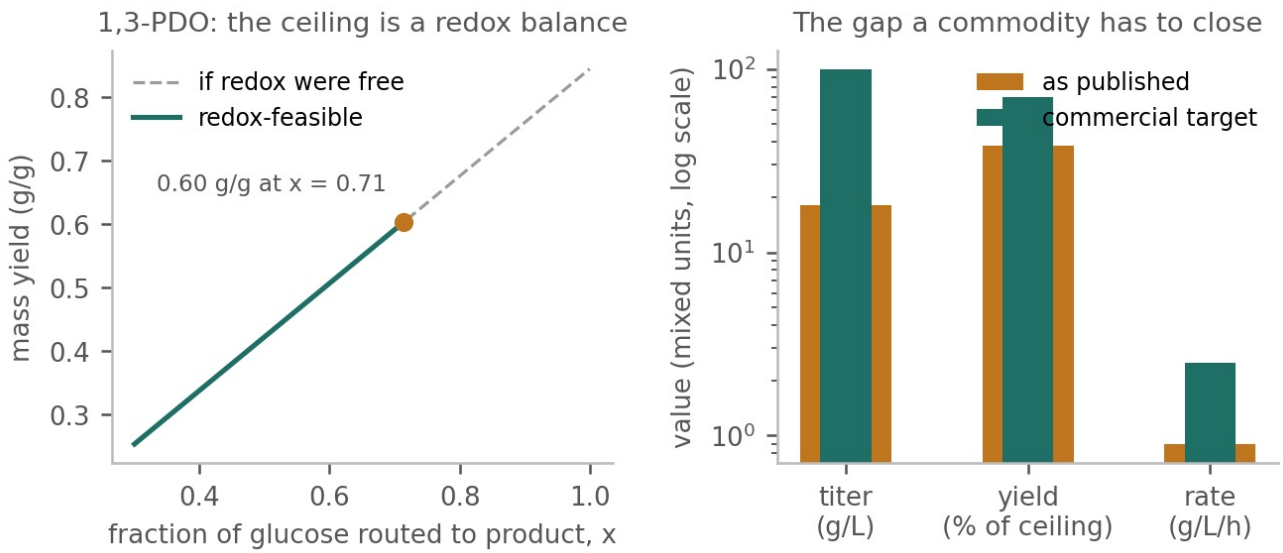


Figure 54.1 Left: the 1,3-propanediol ceiling as a redox balance — the dashed line is what carbon alone would allow, the solid line what redox permits, and the optimum sits where the NADH demand of the product exactly consumes the NADH from oxidising the rest. Right: the gap a commodity has to close, using the reported 1,4-butanediol numbers. Computed from the balances in the text; the right-hand panel's commercial figures are as reported rather than derived.

That is also why the published process re-engineered sugar uptake. The phosphotransferase system spends one phosphoenolpyruvate per glucose imported, and PEP is downstream of the branch point — so importing sugar by the native route burns carbon that the product cannot use, and replacing it with a permease plus a kinase changes the energy balance in the product's favour. Part 1 chapter 7 flagged this for aromatics; here it is the same lever for a different reason, and it is the kind of detail that distinguishes a process built by people who did the balance.

1,4-Butanediol: a molecule no organism makes

Yim and colleagues designed a route to BDO computationally — searching reaction databases for paths from central metabolism to a compound with no natural biosynthesis — then built it in *E. coli*, including an engineered oxidative TCA branch running under anaerobic conditions to supply the precursor and the reducing power. Reread it now with parts 2 and 4 behind you and three things stand out that are invisible on a first reading.

First, the knockouts were chosen by the strain-design algorithms of week 39, not by intuition: the deletions couple product formation to growth so that the strain cannot grow well without making BDO, which converts a screening

problem into a selection. Second, the reducing power comes from running part of the TCA cycle in the oxidative direction without oxygen, which is the kind of thing you only propose after writing the redox balance — it is week 6's anaplerosis argument used in reverse. Third, the paper reports 18 g/L and commercial operation required several times that, which the tracker quotes as above 100 g/L. That gap is the subject of part 2 chapter 18 and is the normal distance between a demonstration and a plant.

The case, written in the form the tracker asks for: host *E. coli*; target 1,4-butanediol, a non-natural product; pathway computationally designed from succinyl-CoA and α-ketoglutarate; tools genome-scale modelling, strain-design algorithms, growth coupling, multiplexed editing; TRY 18 g/L in the paper against above 100 g/L commercially; bottleneck precursor supply and redox under anaerobic conditions; commercial status produced at scale into polymer markets.

Why *C. glutamicum* owns the amino-acid market

Corynebacterium glutamicum was isolated as a glutamate excreter in the 1950s and has made amino acids at multi-million-tonne scale ever since. The properties that put it there are worth listing because they are what a host comparison in part 6 will be built from.

- **It excretes amino acids natively**, through dedicated exporters. Part 2 chapter 14's arithmetic says a charged product needs an exporter; this organism came with them, which is a larger advantage than any pathway engineering.
- **It tolerates very high product concentrations** — over 100 g/L of lysine — which is a tolerance phenotype nobody has had to engineer.
- **It has no significant overflow metabolism** of the acetate kind under normal industrial feeding, so the week 6 problem that dominates *E. coli* fed-batch is much reduced.
- **It is Generally Recognised As Safe in practice**, with decades of regulatory history for food and feed products, which matters enormously for amino acids destined for animal feed.
- **Its genetics are now routine**: genome sequence, strong promoter libraries, CRISPR tools and well-characterised integration sites, which was the missing piece thirty years ago.

The engineering story of lysine in this host is also the best single illustration of parts 1 and 2 working together. Part 1 chapter 3 said the first committed enzyme of an amino-acid pathway is feedback-inhibited by its end product and that a feedback-resistant allele is the single most effective change — in lysine that is aspartokinase, and the resistant allele is the foundation of every production strain. Part 1 chapter 5 computed the ceiling: 0.81 mol per mol, 0.66 g/g, limited not by carbon but by the NADPH the pathway demands and the carbon the pentose phosphate pathway burns to supply it. And the published systems-engineering strains report titers above 100 g/L at yields near half a gram per gram — roughly three quarters of the ceiling — assembled from a dozen defined genomic changes rather than from mutagenesis, which is the field's clearest demonstration that rational design works when the biology is known well enough.

Bioplastics, as the bridge to chapter 56

Cupriavidus necator accumulates polyhydroxybutyrate as a carbon and energy store when nitrogen or phosphorus runs out while carbon keeps arriving — up to the majority of its dry weight. That is a different kind of product from everything else in this chapter: it is intracellular, it is a polymer rather than a molecule, and it is made by deliberately stopping growth. Chapter 56 works it all the way through, because it is the case study with the most to teach about reading a fermentation, and because partitioning carbon between storage polymer and growth is a question the rest of this module has prepared you for exactly.

Constraint sheet — commodity chemicals and amino acids

Binding metric	Yield, then titer, then rate. Feedstock is usually the largest cost line; part 4 chapter 49's model says a strain below about 0.35 g/g cannot clear a $1.80/kg market whatever its titer.
Yield ceiling	High, and usually set by redox rather than by carbon: 0.60 g/g for PDO, 0.66 for lysine, 0.51 for ethanol. Write the NADH balance before the carbon balance.
Titer target	50–150 g/L, because broth volume per kilogram sets separation cost.
Pathway difficulty	Low — two to six genes. The difficulty is in balancing them and in the host, not in finding them.
Host requirements	Fast growth, cheap minimal medium, high tolerance, native export of a charged product, and a regulatory history if the product goes into food or feed.
Characteristic lever	Feedback-resistant first committed enzyme; sugar-uptake re-engineering when PEP is a precursor; growth coupling so that selection replaces screening.
Where it fails	Against an incumbent petrochemical plant with amortised capital and a volatile feedstock price; and in the decade between 18 g/L and 100 g/L.

Three things people get wrong

- ‘The ceiling is set by carbon.’ For most commodity routes it is set by redox — PDO's 0.60 g/g is a NADH balance, not a carbon count.
- ‘Non-natural products are the hard ones.’ BDO was designed and built in a few years; closing the titer gap took far longer.
- ‘*C. glutamicum* is a legacy host.’ Its advantages — native export, tolerance, little overflow, regulatory history — are exactly the ones that are hardest to engineer into something else.

Self-check — answer on paper first

1. Derive the 1,3-propanediol yield ceiling and explain in one sentence why the number is below the carbon limit.

Two PDO per glucose needs four NADH and the route to PDO generates none, since glycolysis makes NADH only below the branch point, so part of the feedstock must be fully oxidised to supply reductant: with roughly 10 NADH per glucose oxidised, $4x = 10(1 - x)$ gives $x = 0.71$ and 1.43 mol per mol, or 0.60 g/g. It is below the carbon limit of 2 mol per mol because the product is more reduced than the substrate per carbon, so electrons, not carbon atoms, are the scarce resource.

2. Why does replacing the phosphotransferase sugar-uptake system help a PDO strain, and what does it cost?

Because the phosphotransferase system consumes one phosphoenolpyruvate per glucose imported, and PEP lies below the branch point to glycerol, so the energy for uptake is taken from carbon the product cannot use — effectively a tax on every glucose. Replacing it with a permease plus glucokinase pays the tax in ATP instead, which the oxidative part of the metabolism can supply more cheaply. The costs are a slower uptake rate, which can limit productivity, and a loss of the regulatory coupling the native system provides.

3. Name the three properties of *C. glutamicum* that would be hardest to engineer into *E. coli*, and say why.

Native export of amino acids, because finding or engineering a specific exporter is the problem part 3 chapter 45 showed to be a screening exercise with a poor base rate; tolerance to above 100 g/L of product, because tolerance is polygenic and usually evolved rather than designed; and regulatory acceptance for food and feed, which is not a biological property at all and cannot be engineered on any timescale. The absence of acetate overflow is a fourth, and is the one that is most nearly addressable by engineering.

What to open a book for. Yim et al., Nat. Chem. Biol. 7:445 (2011), reread with parts 2 and 4 behind you; the EPA's 2003 Greener Reaction Conditions award summary for bio-PDO, which is short and unusually clear about why the process was built the way it was; and a *C. glutamicum* systems-metabolic-engineering review for the lysine strain lineage. Stephanopoulos chapters 2 and 3 remain the reference for the yield arithmetic.

Biofuels, C1 and gas fermentation, and food ingredients

Three regimes in one week, united by a single question: what is the feedstock, and what does getting it into the cell cost? For fuels the answer killed an industry; for gas it rebuilt one.

Higher alcohols from 2-ketoacids: one idea, a family of products

Atsumi, Hanai and Liao's non-fermentative route is the most elegant piece of pathway design in the fuels literature, and its logic is worth being able to draw from memory. Every amino acid is made from a 2-ketoacid, and every 2-ketoacid can be decarboxylated and then reduced — the Ehrlich degradation that yeast uses to make fusel alcohols — to give an alcohol one carbon shorter. So the amino-acid biosynthetic network, which cells already run at high flux, becomes a precursor network for alcohols: 2-ketoisovalerate from the valine pathway gives isobutanol, 2-keto-3-methylvalerate gives 2-methyl-1-butanol, phenylpyruvate gives 2-phenylethanol, and so on. Two enzymes — a broad-specificity ketoacid decarboxylase and an alcohol dehydrogenase — convert the whole family.

The reported isobutanol result was 22 g/L at 86% of theoretical. Check the theoretical: isobutanol is $C_4H_{10}O$, one mole per mole of glucose with two CO_2 lost, so 74.12/180.16 = **0.411 g/g**, as part 2 chapter 18 computed. Eighty-six per cent of that is a remarkable yield for a first demonstration and reflects the fact that the route is nearly redox-balanced on glucose, which is why it works anaerobically.

Two refinements the week asks you to understand. The route consumes NADPH in the ketol-acid reductoisomerase step while glycolysis makes NADH — part 2 chapter 19's cofactor mismatch, and the standard fix is the cofactor swap, which in this pathway has been done. And in yeast, moving the pathway into the mitochondrion raised titers: the precursor 2-ketoisovalerate is made there, so compartmentalising the downstream enzymes with it avoids exporting an intermediate and keeps the local concentration high — the same argument as part 4's compartmentalisation note, with a measured payoff.

Why advanced biofuels mostly did not happen

The honest account, which you should be able to give without sounding either bitter or naive. The technical programmes largely worked: isobutanol, farnesene, fatty alcohols and renewable diesel were all demonstrated at pilot scale by companies built around them. What killed or slowed them was arithmetic external to the biology. Fuels are the cheapest products by mass that anyone makes, so the margin per kilogram is cents; the incumbent is an oil price that fell by half in 2014 and has swung since; the capital for a fuel-scale plant is hundreds of millions, which must be raised against that price; and the yield ceiling of a reduced product on sugar is fixed by stoichiometry at a level that leaves little room.

Part 4 chapter 49's model makes the point precisely: at a product price near $1/kg, the feedstock term alone at a realistic yield consumes most of it, and there is no titer that fixes that. The companies that survived pivoted to higher-value products made on the same platforms — farnesene to cosmetics and flavours rather than diesel, for instance — which is not a failure of the science but a correct response to a cost structure. The lesson for your own target selection is the one in chapter 53: choose products whose incumbent cannot halve its price.

Gas fermentation: the acetogens and why ATP is so tight

Gas fermentation is the exception to the pessimism above, and it is instructive that the reason is feedstock rather than biology. *Clostridium autoethanogenum* and relatives grow on carbon monoxide, CO_2 and hydrogen through the Wood–Ljungdahl pathway, and LanzaTech's first commercial plant started in 2018 converting steel-mill off-gas to ethanol. The feedstock is a waste stream with a negative or near-zero price, which changes the whole cost structure.

The energetics are the thing to understand, because they explain the product range. Wood–Ljungdahl is the most ATP-efficient route to acetyl-CoA known, and that is faint praise: it is very nearly ATP-neutral, with acetogens living on fractions of an ATP per acetate formed, conserving energy through a membrane ion gradient generated by ferredoxin oxidation and flavin-based electron bifurcation rather than through substrate-level phosphorylation. A cell with almost no ATP margin cannot afford long biosynthetic routes, which is why the natural products are acetate and ethanol, why engineered products are mostly two- to four-carbon compounds close to acetyl-CoA, and why anything requiring significant ATP must have an external energy input.

Carbon efficiency follows from the chemistry of the feedstock, and the two cases differ dramatically:

```
6 CO + 3 H₂O  →  C₂H₅OH + 4 CO₂        only 33% of the carbon reaches the product
2 CO₂ + 6 H₂  →  C₂H₅OH + 3 H₂O        all of the carbon reaches the product
```

On CO alone, two thirds of the carbon is sacrificed to supply electrons, because CO is both the carbon source and the reductant. Add hydrogen as a separate electron donor and the carbon is fully conserved — which is why the field's trajectory is toward CO_2 plus H_2, and why chapter 57 is about hydrogen and electricity rather than about gas.

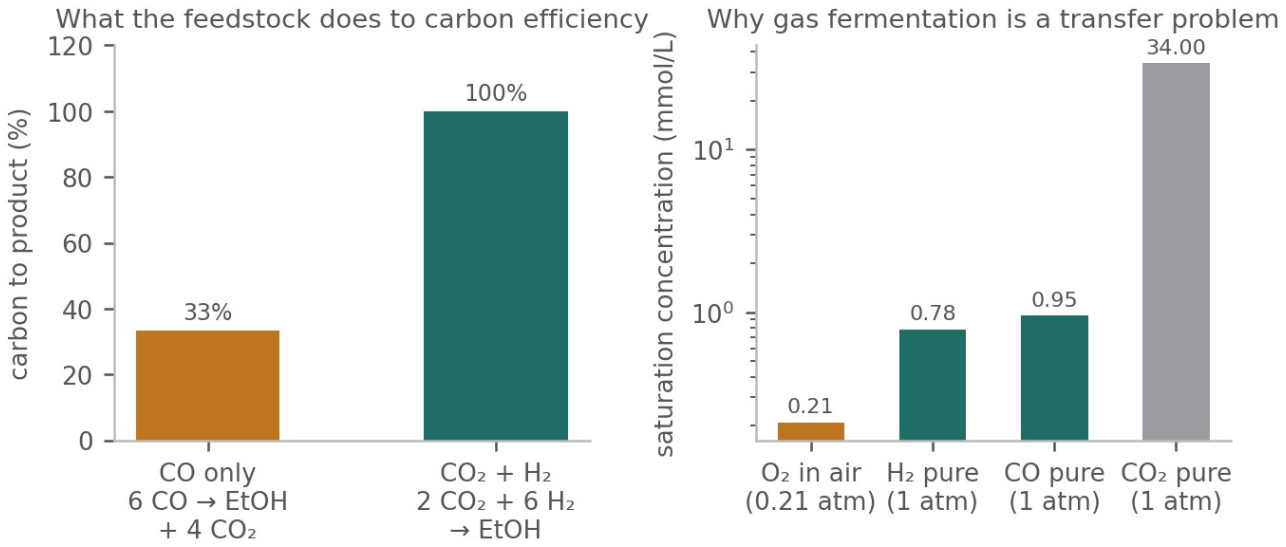


Figure 55.1 Left: carbon efficiency of the two gas-fermentation stoichiometries, from the balanced equations above. Right: saturation concentrations of the gases involved, on a log scale — the reason gas fermentation is a mass-transfer discipline. Values are standard Henry's-law figures at ambient conditions and are approximate; the comparison, not the third digit, is the point.

And the operational consequence, which is figure 55.1's right-hand panel: CO and H_2 are about as soluble as oxygen, which is to say barely, while CO_2 is two orders of magnitude more soluble. So a gas fermentation is limited by gas–liquid transfer of the sparingly soluble components, exactly like an aerobic fermentation but with two or three such components at once, and the reactor designs that dominate — loop reactors, bubble columns with long gas residence times — are chosen for transfer rather than for mixing.

C1 substrates, compared honestly with sugar

Methanol and formate are the other one-carbon feedstocks under development, and the comparison with sugar is worth having at hand.

Feedstock	Advantage	Cost in engineering
Glucose	Universal metabolism, high solubility, cheap at scale, no transfer problem	Competes with food and land; the price is volatile
Methanol	Liquid, fully miscible, made from natural gas or CO_2, more reduced than sugar so higher yields for reduced products	Toxic above a few per cent; native methylotrophs have modest tools; synthetic methylotrophy in *E. coli* or yeast is recent and slow-growing
Formate	Liquid, electrochemically made from CO_2, easy to transport, no transfer problem	Carries only one electron pair per carbon, so ATP and reductant are tight; toxic as formic acid; few hosts use it natively for growth
CO / CO_2 + H_2	Waste or atmospheric carbon; full carbon conservation with H_2	Gas transfer, very tight ATP, safety, and the hydrogen has to come from somewhere — chapter 57

The honest summary is that the one-carbon routes are not yet competitive with sugar for most products, and that their case rests on what happens to the price of renewable electricity and on whether carbon accounting is priced. Which hosts are being developed matters for part 6: methylotrophic yeasts and *Bacillus methanolicus* for methanol, engineered *E. coli* and *Cupriavidus* for formate and H_2/CO_2, acetogenic clostridia for CO.

Food ingredients, and why the host choice was a yeast

Soy leghemoglobin, the molecule that gives a plant-based burger its colour and flavour, is produced in *Komagataella phaffii* — the yeast formerly called *Pichia pastoris* — and was cleared through the FDA's GRAS notification route, notice GRN 737. The host choice is a teaching case in itself, and the week asks exactly why a yeast rather than a bacterium.

Four reasons, each from an earlier part of this module. It is a secreted protein, and *K. phaffii* secretes at grams per litre with a clean background, which is part 1 chapter 1's cell-volume argument: a secreted product has no intracellular ceiling. It is a haem protein, so the host must supply haem at a stoichiometry of one per molecule of a protein made at high titer, and the engineering of haem supply alongside the apoprotein is the core of the process. It is a eukaryotic protein, so folding and any modifications are more reliable in a eukaryote, and the reducing cytoplasm problem of part 1 chapter 2 does not arise for a secreted route. And regulatory acceptance: the host has a long history in food enzymes, which shortens the GRAS argument substantially, and as chapter 54 noted, regulatory history is the one host property that cannot be engineered.

Precision fermentation for food is now the field's fastest-growing commercial segment, for a reason that follows from chapter 53's lesson: the incumbent — an animal or a crop — cannot halve its price in response, and the product's value per kilogram sits comfortably above a commodity chemical's.

Constraint sheet — fuels, gas and food

Fuels: binding metric	Yield, and it is not enough. At cents-per-kilogram margins, the feedstock term alone consumes the price; no titer fixes it.
Fuels: ceiling	0.41 g/g isobutanol, 0.51 ethanol — fixed by the stoichiometry of a reduced product on sugar.
Gas: binding metric	Gas–liquid transfer and ATP. Carbon efficiency is 33% on CO alone and 100% with H_2 as the electron donor.
Gas: host requirement	An acetogen or a knallgas organism; products must sit close to acetyl-CoA because there is almost no ATP margin.
Food: binding metric	Regulatory clearance, purity and consistency; titer matters mainly through cost of goods.
Food: host requirement	Secretion at grams per litre, eukaryotic folding, cofactor supply (haem) at product stoichiometry, and an existing food-use history.
Where each fails	Fuels: the oil price. Gas: transfer rates, safety and the cost of hydrogen. Food: consumer acceptance and the length of a novel-food filing.

Three things people get wrong

- ‘Biofuels failed because the science didn't work.’ The science largely worked; the margin per kilogram and the oil price did not.
- ‘Gas fermentation is just anaerobic fermentation on a different substrate.’ It runs on a near-zero ATP margin and is limited by gas transfer, which constrains the product range.
- ‘CO_2 fixation means carbon-neutral.’ On CO alone, two thirds of the carbon leaves as CO_2; with H_2, the hydrogen's own footprint becomes the question.

Self-check — answer on paper first

1. Explain why one decarboxylase and one dehydrogenase give access to a whole family of alcohols.

Because every amino acid derives from a 2-ketoacid, and the Ehrlich route — decarboxylation then reduction — converts any 2-ketoacid into the corresponding alcohol with one carbon fewer. Both enzymes have broad specificity, so the product is set by which 2-ketoacid accumulates, which is set by which amino-acid branch is amplified. The engineering therefore moves from the two terminal enzymes to the precursor-supply branch, which is where the real work is.

2. Why does adding hydrogen to a CO-fed gas fermentation triple the carbon efficiency?

Because on CO alone the same molecule serves as carbon source and electron donor, so two of every three carbons must be oxidised to CO_2 to supply the electrons for reducing the third — the balanced equation is 6 CO → ethanol + 4 CO_2. Hydrogen supplies electrons without carbon, so the carbon can all be fixed: 2 CO_2 + 6 H_2 → ethanol + 3 H_2O. The cost moves from carbon to hydrogen, and therefore to electricity, which is chapter 57's subject.

3. A collaborator proposes making a 20-carbon product in an acetogen on CO_2 and H_2. What is your first objection?

ATP. Acetogens conserve barely a fraction of an ATP per acetate through a chemiosmotic mechanism, so there is almost no energy margin for a long biosynthetic route — a C20 product requires many ATP-consuming condensation and reduction steps and would not leave enough for maintenance, let alone growth. The options are an external energy input, a two-stage process in which a different organism takes the acetate the acetogen makes, or a different host with respiratory energy conservation such as a knallgas bacterium growing on H_2 and O_2, which is what chapter 57 uses.

What to open a book for. Atsumi, Hanai & Liao, Nature 451:86 (2008) for the 2-ketoacid route; a recent gas-fermentation or acetogen review for the energetics, since the quantitative account of electron bifurcation is newer than any textbook; and FDA GRAS notice GRN 737 for soy leghemoglobin, which is worth reading as an example of what a regulatory filing actually contains — it is not a biology document and that is the point.

PHA, from pathway to polymer: the case study worked all the way through

Three enzymes, one glucose, one monomer. This is the simplest pathway in the module and the richest case study, because everything that decides whether it works sits outside the pathway: the redox surplus, the nutrient feed, and the difference between content and yield.

The pathway, and its balance

Draw it once and you will not forget it. Glucose through glycolysis to two pyruvate, each oxidised by pyruvate dehydrogenase to acetyl-CoA. Then three enzymes: **PhaA**, a β-ketothiolase, condenses two acetyl-CoA to acetoacetyl-CoA; **PhaB**, an NADPH-dependent reductase, reduces it to (R)-3-hydroxybutyryl-CoA; **PhaC**, the synthase, polymerises it and releases coenzyme A. Mark every cofactor as you draw, and the net equation per monomer unit is

```
glucose + 1 NADPH  →  (3-hydroxybutyrate unit) + 2 CO₂ + 4 NADH + 2 ATP + 1 NADPH consumed

carbon: 4 of 6 reach the polymer (67%)        mass yield ceiling: 86.09/180.16 = 0.478 g/g
cofactors: +2 NADH (glycolysis) +2 NADH (PDH) −1 NADPH  →  a surplus of 3 reduced cofactors
```

That surplus is the thing to carry out of this week. The polymer is slightly *less* reduced per carbon than glucose, so making it leaves the cell with electrons it must dispose of, and the only disposal route is respiration: three reduced cofactors per glucose is 1.5 O_2 per monomer unit, which is **0.56 g of oxygen per gram of PHB**, or about 35 $mmol \cdot L^{-1} \cdot h^{-1}$ of oxygen demand at a productivity of 2 g/L/h, on top of whatever the residual biomass requires for maintenance. PHB production is therefore an aerobic process with a real oxygen duty, and oxygen transfer is one of the two things that limit it at scale.

Why can a strain not reach 0.478 g/g even with perfect enzymes? Two reasons, both structural. Carbon must also build the residual biomass — the catalytic cells doing the work — and that carbon never becomes polymer. And the cell must pay maintenance throughout the accumulation phase, which in a non-growing culture is met by oxidising substrate to CO_2 with no product at all. Both are substrate consumed for something other than polymer, so the achievable yield is the ceiling minus the biomass investment minus maintenance, and the longer the accumulation phase runs, the larger the second term grows.

The hosts, and what each one costs you

Host	What it gives	What is harder than in *E. coli*
Cupriavidus necator	The native short-chain-length pathway, very high accumulation (majority of dry weight), tolerance of nutrient limitation, and autotrophic growth on $H_2 + CO_2 + O_2$ — which is chapter 57	Slower growth, conjugation rather than simple transformation for DNA delivery, fewer characterised promoters and integration sites, and restriction barriers in wild isolates (part 4 chapter 52)
Pseudomonas putida	Medium-chain-length PHA from β-oxidation of fatty acids or de novo fatty-acid synthesis, giving elastomeric polymers; robust to solvents and aromatics	mcl-PHA is a mixture of monomers rather than a defined polymer; the carbon source is usually fatty acids, which are more expensive than sugar
Recombinant *E. coli*	Every genetic tool in the module, fast growth, no native depolymerase so the polymer is not re-consumed, and high content from a three-gene operon	No native accumulation physiology, so the nutrient-limitation trigger has to be imposed by the process; lower tolerance of the very high cell densities the process wants

Note the pattern, because it recurs in part 6: the host with the best native physiology has the worst genetic tools, and the host with the best tools has no native physiology. Which to choose depends on how much of the process you intend to engineer and how much you intend to inherit.

Reading a fed-batch run, which is the real skill

Here is a run. Six time points, and the only measurements are total dry cell weight, PHB content as a percentage of that weight, and cumulative glucose consumed — which is all most papers give you.

Time (h)	0	12	24	36	48	60
Dry cell weight (g/L)	1.0	15	45	70	95	110
PHB content (% of DCW)	0	5	20	45	62	72

Glucose consumed (g/L)	4	40	110	190	280	350
PHB (g/L)	0	0.8	9.0	31.5	58.9	**79.2**
Residual biomass (g/L)	1.0	14.2	36.0	38.5	36.1	**30.8**

Now the four numbers that matter, none of which the table gives directly.

- **Residual biomass** — dry weight minus polymer — is the catalyst, and it plateaus at 36–38 g/L around 24 to 36 h and then declines slightly. That plateau is the nutrient feed doing its job: nitrogen ran out, growth stopped, and carbon kept flowing into polymer. The small decline afterward is maintenance consuming cell material. If residual biomass had kept rising, the limitation had not been imposed and you would be making biomass instead of polymer; if it had collapsed, the cells were dying and the run should have been harvested earlier.
- **Volumetric productivity** is 79.2/60 = **1.32 g/L/h** over the whole run, and (79.2 − 9.0)/36 = **1.95 g/L/h** over the accumulation phase alone. Quote the first; the second is the one papers quote, and the difference is the growth phase, as part 2 chapter 18 warned.
- **Overall yield** is 79.2/350 = **0.226 g/g**, which is 47% of the ceiling — and that is the honest process number, because the growth-phase glucose was paid for too.
- **Accumulation-phase yield** is (79.2 − 9.0)/(350 − 110) = **0.293 g/g**, 61% of the ceiling. The remaining 39% is the gap worth decomposing: maintenance of 36 g/L of cells over 36 hours, the CO_2 from the two decarboxylations, and any residual growth.

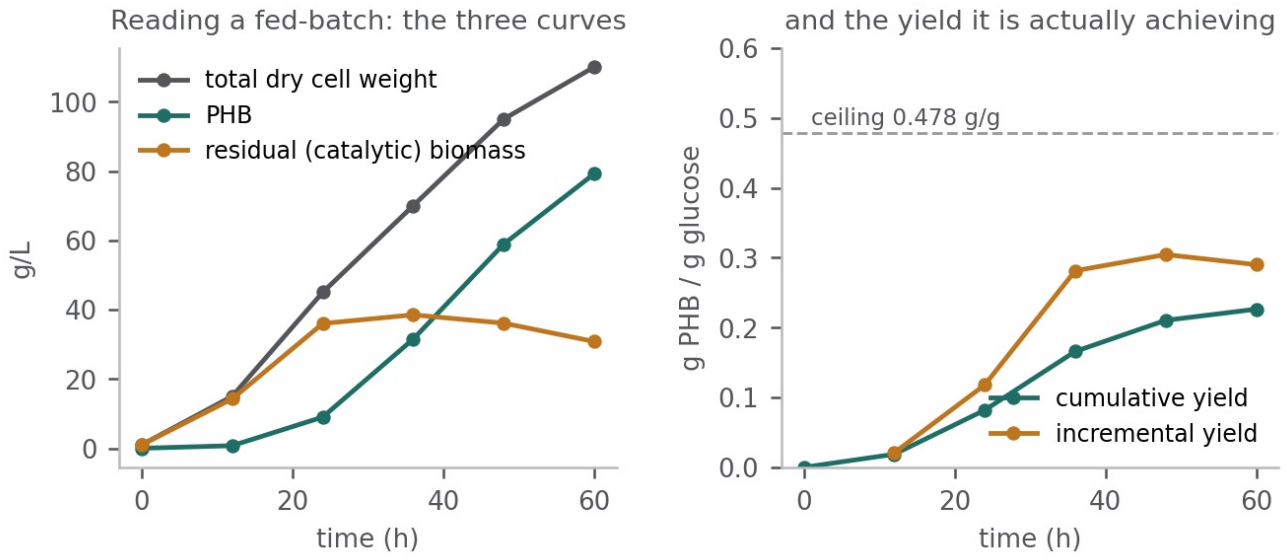


Figure 56.1 Left: the three curves to extract from any PHA fed-batch table — total dry weight, polymer, and residual catalytic biomass, whose plateau marks the nutrient limitation. Right: cumulative and incremental yield against the 0.478 g/g ceiling; the incremental yield peaks at 0.30 g/g around 48 h and then falls as maintenance consumes a larger share. Computed from the table above, which is constructed for this module rather than taken from a paper.

Look at the incremental yield in the right-hand panel: 0.28 between 24 and 36 h, 0.30 between 36 and 48 h, and 0.29 thereafter — it is already turning down by the last interval. That is the signal to harvest, and it is invisible in the cumulative number, which is still rising. **Incremental yield falling below the point where the marginal glucose pays for itself is the harvest criterion**, and computing it from the table is the single most useful thing you can do with someone else's fermentation data.

What to change in the next run, from these numbers alone: the growth phase consumed 110 g/L of glucose to build 36 g/L of residual biomass, which is nearly a third of the total substrate for catalyst rather than product, so a shorter growth phase to a lower residual biomass — or a richer inoculum — would raise the overall yield even if it lowered the peak productivity. And since the incremental yield turns down by 60 h, harvesting at 48–54 h loses a little content and gains yield and productivity. The measurement that tells you it worked is the incremental yield curve, not the final content.

Copolymers: why composition is the product specification

Poly(3-hydroxybutyrate) alone is a stiff, brittle polymer with a melting point uncomfortably close to its degradation temperature, which is the reason PHB has never displaced polypropylene on its own. Incorporating a second monomer fixes it: poly(3-hydroxybutyrate-*co*-3-hydroxyvalerate) with a few mole per cent of the C5 unit has a lower melting point, a wider processing window and much better toughness. So for this product class **the specification is a composition,**

not a titer — a strain making 80 g/L of the wrong copolymer fraction has made the wrong product.

The engineering handle is the supply of the C5 precursor, propionyl-CoA. Classically that meant feeding propionate or valerate, which is expensive and somewhat toxic. The more interesting route makes it from sugar: 2-ketobutyrate, from the threonine pathway or from citramalate, is the entry point, and Jo and colleagues showed that overexpressing acetolactate synthase in *C. necator* H16 modulates the 3-hydroxyvalerate fraction from unrelated carbon — which is precisely the kind of result that only makes sense if you can see the amino-acid network as a precursor network, exactly as chapter 55's 2-ketoacid route does.

Feedstocks, inhibitors, and what "robust" means

Lignocellulosic hydrolysate is the feedstock everyone wants and few strains tolerate, and the reason is a short list of inhibitors generated by the pretreatment itself. Four to measure, with the assay for each:

- **Furfural**, from pentose dehydration — HPLC with UV detection, or GC-MS.
- **5-hydroxymethylfurfural**, from hexose dehydration — the same HPLC run.
- **Acetic acid**, from deacetylation of hemicellulose, and the one that matters most at neutral-to-low pH for the weak-acid reasons of part 3 chapter 45 — HPLC with refractive index or UV, on the organic-acid column you already run.
- **Phenolics**, from lignin depolymerisation, which are the most variable and the least characterised — total phenolics by a colorimetric assay for a quick number, GC-MS or LC-MS for the individual compounds when it matters.

"Robust", applied to a strain, should mean something measurable: growth rate and yield maintained at a stated concentration of each inhibitor, or in a stated hydrolysate, against the parent. Sun and colleagues' report of efficient PHB production from lignocellulosic hydrolysate with a *C. necator* strain is the kind of claim to read with that definition in hand — what concentrations, which inhibitors, against what control, and whether the improvement was evolved or engineered. Those are the questions a member of that lab would expect you to ask, and asking them is the point of the week's reading.

Recovery and characterisation, and what to trust

PHA content and monomer composition are measured by gas chromatography after methanolysis — the polymer is digested into its methyl esters, which are quantified against standards, with benzoic acid or a similar compound as internal standard. Molecular weight comes from gel permeation chromatography, thermal properties from differential scanning calorimetry, and composition can be confirmed by NMR. Recovery from the cells is by solvent extraction, typically chlorinated, or by digesting the non-polymer biomass with hypochlorite or enzymes — and the choice matters for the molecular weight, because aggressive digestion cleaves chains.

Now connect this to part 4 chapter 50, because a PHA titer is a long chain of multiplications: a GC peak area, a calibration against a methyl ester standard, a methanolysis efficiency that is assumed rather than measured, a dry-weight determination, and a content percentage derived from both. Before trusting a reported titer I would want: the calibration range and whether it was weighted; the methanolysis recovery measured on a commercial polymer standard of known mass; the dry weight method; and the agreement between content × dry weight and an independent gravimetric extraction. A content figure above 90% should prompt the question of whether the residual biomass was measured or inferred, since at that level a small dry-weight error swings the content by several points.

Constraint sheet — PHA and intracellular polymers

Binding metrics	Three that trade against each other: polymer content (% of dry weight), volumetric productivity, and yield on substrate. A strain is only good if all three are.
Yield ceiling	0.478 g/g on glucose, 67% of carbon; unreachable because residual biomass and maintenance consume substrate without making polymer.
Cofactor balance	A surplus of three reduced cofactors per glucose — this is an oxygen duty of 0.56 g O_2 per g PHB, not a redox deficit.
Process requirement	Two phases: growth, then nutrient limitation with carbon still fed. Residual biomass must plateau; if it rises, the limitation was not imposed.
Product specification	Composition, not just quantity: the 3HV fraction sets the material properties, and the engineering handle is propionyl-CoA supply.
Host requirements	Native accumulation physiology and tolerance of nutrient limitation (*C. necator*), or full genetic tooling and an imposed process (*E. coli*). No host gives both.
Where it fails	Against polypropylene on price, unless the feedstock is waste or the polymer's biodegradability is valued; and on molecular weight after aggressive recovery.

Three things people get wrong

- ‘High PHB content means a good strain.’ Content is one of three metrics; a 90% content at 0.15 g/g and 0.3 g/L/h is a poor process.
- ‘PHB production needs reducing power.’ It produces a surplus of three reduced cofactors per glucose, so what it needs is oxygen.
- ‘The copolymer is a variant of the same product.’ The 3HV fraction is the specification; without it the polymer is too brittle to sell.

Self-check — answer on paper first

1. Write the glucose-to-PHB net equation from memory with its cofactors, and state the oxygen demand per gram of polymer.

Glucose → 2 pyruvate (2 ATP, 2 NADH) → 2 acetyl-CoA (2 NADH, 2 CO_2); PhaA condenses them, PhaB reduces with 1 NADPH, PhaC polymerises. Net per monomer: one glucose in, one C4 unit plus 2 CO_2 out, 4 NADH and 2 ATP generated, 1 NADPH consumed — a surplus of three reduced cofactors. Clearing that surplus by respiration takes 1.5 mol O_2 per mol of monomer, which is 48 g O_2 per 86 g of polymer, or 0.56 g of oxygen per gram of PHB.

2. In the table above, residual biomass declines from 38.5 to 30.8 g/L between 36 and 60 h. What does that mean and does it matter?

It means the catalytic cells are consuming their own protein and other components to meet maintenance, which is what happens when nitrogen has been withheld for a long accumulation phase. It matters in two ways: the specific productivity of the remaining cells sets the volumetric rate, so a shrinking catalyst means a falling rate even if content keeps rising, and the material consumed was built from substrate you paid for, so it depresses the overall yield. Combined with the incremental yield turning down, it is the argument for harvesting earlier.

3. A paper reports 92% PHB content and 0.40 g/g yield in *C. necator* on glucose. What is your reaction?

The content is plausible for this organism; the yield is 84% of the stoichiometric ceiling of 0.478, which is implausibly high once residual biomass and maintenance are accounted for, so I would check how the yield was computed. The likely explanations are that it is an accumulation-phase yield rather than an overall one, that the substrate consumed excludes the growth phase or a co-substrate, or that the dry-weight and content measurements are inconsistent in the direction that flatters both. The test is whether the carbon balance closes: polymer plus residual biomass plus CO_2 should account for the glucose, and at 0.40 g/g overall there is very little room for the CO_2 the pathway must release.

What to open a book for. No book. A *C. necator* PHA review for the hosts and physiology; Sun et al., Green Chemistry (2026) for the lignocellulosic strain and Jo et al., Int. J. Biol. Macromol. (2023) for the copolymer handle — both from the lab whose recruiting notice names them, so read them as the group would expect, with one question each that a lab member could answer in two sentences. Part 2 chapter 18 and part 4 chapter 50 are the two chapters of this module to have open alongside them.

CO_2, hydrogen and waste carbon: when electricity is the feedstock

A CO_2 route's feedstock is not carbon dioxide, which is nearly free. It is hydrogen, and therefore electricity, and therefore a number in kilowatt-hours per kilogram that decides everything.

The stoichiometric floor, and what it leaves out

Start with the balance the week asks you to write, for a 3-hydroxybutyrate unit made from CO_2 and hydrogen:

```
4 CO₂ + 9 H₂  →  C₄H₆O₂ + 6 H₂O
carbon 4 = 4 · hydrogen 18 = 6 + 12 · oxygen 8 = 2 + 6        the equation balances
```

Nine hydrogen per monomer unit, which is 105 mol of H_2 per kilogram of polymer. This is a floor and not an estimate, and it is important to understand why: it counts **reducing equivalents only**. It assumes the cell spends no energy fixing the carbon, which is false — the Calvin cycle costs ATP — so the real demand is higher by however much ATP the organism must make, and ATP in a knallgas organism is made by respiring hydrogen with oxygen.

Now tally the pathway, which is the week's second exercise and produces a satisfying check. Fixing six CO_2 to one hexose by the Calvin cycle costs 18 ATP and 12 NADPH. Glycolysis and pyruvate dehydrogenase then return 2 ATP and 4 NADH while releasing 2 CO_2, and PhaB consumes 1 NADPH. Net per monomer unit:

```
16 ATP   and   13 NADPH − 4 NADH = 9 net reducing equivalents
```

Nine — exactly the floor. The stoichiometric equation and the pathway tally agree on reductant, as they must, and the pathway reveals what the equation hid: **16 ATP that the equation does not count**. Supplying those by respiring hydrogen adds 16/(ATP per H_2) more hydrogen, and that assumption is the one that moves the answer most:

ATP per H_2 oxidised	Extra H_2 for ATP	Total H_2 per unit	mol H_2 per kg PHB	Electrolysis energy
1.0	16.0	25.0	290	31.9 kWh/kg
1.5	10.7	19.7	228	25.1 kWh/kg
2.0	8.0	17.0	197	21.7 kWh/kg
3.0	5.3	14.3	166	18.3 kWh/kg

Electrolysis taken at 0.11 kWh per mole of H_2, roughly 55 kWh/kg of hydrogen, which is a reasonable figure for alkaline or PEM electrolysis including balance of plant. The ATP yield per H_2 is the organism's P/O ratio question from part 1 chapter 4, and the table is the quantitative reason that chapter insisted you treat it as an assumption rather than a constant.

So the energy demand sits between roughly 18 and 32 kWh per kilogram of polymer depending on a parameter you cannot look up for your organism. Nearly a factor of two, from one assumption. This is the honest state of any CO_2-route calculation, and stating the range with its cause is a much stronger position than quoting a single number.

Electricity as the real feedstock, and the break-even price

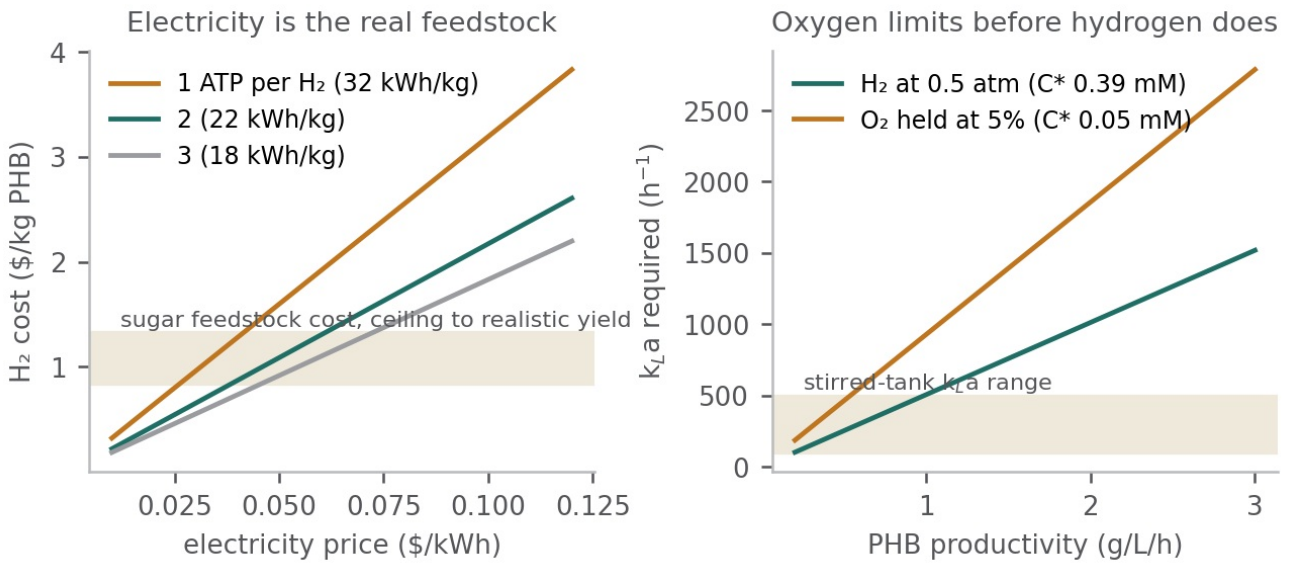


Figure 57.1 Left: hydrogen cost per kilogram of PHB against electricity price for three ATP-yield assumptions, with the sugar route's feedstock cost band — $0.84/kg at the stoichiometric ceiling to $1.33/kg at a realistic 0.30 g/g — shaded. Right: the gas-transfer coefficient required for a target productivity, showing that oxygen, held low for safety, limits before hydrogen does. Computed from the balances in this chapter; electrolysis energy, gas solubilities and the safety-limited oxygen fraction are the stated assumptions.

At 2 ATP per H_2 — 21.7 kWh/kg — electricity at $0.03/kWh puts the hydrogen cost at $0.65/kg of polymer, $0.05/kWh at $1.09, and $0.10/kWh at $2.17. Against the sugar route's feedstock cost of $0.84/kg at the ceiling and $1.33/kg at a realistic yield, the break-even electricity price is about **$0.04/kWh against a perfect sugar process and about $0.06/kWh against a real one**. Those are prices that exist today in a few places with curtailed renewable generation, and nowhere near the average industrial tariff.

What that number leaves out is as important as the number. Electrolyser capital, which at the scale required is a plant in its own right. CO_2 capture and supply, which is free only when co-located with a concentrated source. Gas handling, compression and recycle. Downstream processing, which is identical to the sugar route's and therefore does not favour either. And the safety engineering below. In part 4 chapter 49's framework, the hydrogen cost enters as the feedstock line and the electrolyser as capital, and the right way to present a CO_2 route is in that model rather than as a kilowatt-hour figure — because the kilowatt-hour figure alone makes the route look better than it is.

Moving the gas into the liquid, which is where it actually fails

Part 3 chapter 48's lesson applies again, with hydrogen added to oxygen. At a target of 2 g/L/h of PHB and 17 H_2 per monomer, hydrogen demand is 395 mmol·L^{-1}·h^{-1}, and the ATP-related hydrogen must be respired with oxygen at 93 mmol·L^{-1}·h^{-1}. Required transfer coefficients follow from $N = k_La \cdot C^*$:

```
H₂ at 0.5 atm partial pressure, C* ≈ 0.39 mM  →  k_L a ≈ 1,000 h⁻¹
O₂ held at 5% of the gas phase, C* ≈ 0.05 mM  →  k_L a ≈ 1,900 h⁻¹
```

A well-agitated stirred tank reaches a few hundred per hour. Both numbers are therefore out of reach at that productivity, and **oxygen is the binding one** — not because it is needed in larger quantity, it is not, but because its concentration in the gas phase must be kept low for safety, which cuts its driving force by a factor of four relative to air. That is the counter-intuitive result of the week and it is worth being able to state: in a hydrogen-oxidising fermentation, the gas that limits is the one you are deliberately starving the system of.

What raises k_La, and what each costs: more power input per volume, which costs electricity and shears the cells; smaller bubbles through sparger design or a loop reactor, which costs pressure drop; higher pressure, which raises C* proportionally and costs vessel capital and compression — and is the lever gas fermentation companies actually use; and gas recycle with a high recirculation rate, which costs compression. The design answer is usually a pressurised loop reactor rather than a stirred tank, which is why gas-fermentation plants do not look like fermenters.

Safety, which shapes the design rather than annexing it

Hydrogen and oxygen in the same vessel is a flammable mixture over a very wide composition range, and the standard mitigation is to hold the oxygen fraction below the limiting oxygen concentration — for hydrogen mixtures, around 5%, which is exactly the constraint that made oxygen the transfer bottleneck above. So safety is not a paragraph appended to the proposal; it is the reason the process operates where it does.

The statement to put at the top of any proposal involving these gases, and to know who reviews it: the process operates with hydrogen and oxygen in a single gas phase; the oxygen fraction is controlled below the limiting oxygen concentration with redundant analysers and automatic interlocks; the vessel and gas loop are rated and relieved for the credible overpressure case; and the design is reviewed by the institution's safety committee, with a HAZOP or equivalent, before any gas is connected. In most institutions that review involves the laboratory safety officer, the engineering or facilities group for the gas supply and ventilation, and a formal committee for the pressure system — and it happens before work starts, not alongside it.

Bioelectrochemical and hybrid routes

If electricity is the real feedstock, the obvious question is why make hydrogen at all. Microbial electrosynthesis supplies the electrons directly: cells grow at or near a cathode and take up reducing equivalents from it, either directly or through hydrogen generated in situ at the electrode surface. The architecture is a cathode chamber with the culture and the medium, an ion-exchange membrane, and an anode chamber where water is oxidised — a schematic worth drawing, because the engineering constraints are visible in it.

Two of those constraints are the research frontier. **Current density** limits volumetric productivity directly: electrons per second per square metre of cathode, against a vessel volume, which means electrode area per unit volume becomes the scaling parameter — the same surface-to-volume problem as the gas transfer above, in a different currency. And **biocompatibility** of the catalyst means something specific here: an electrocatalyst for hydrogen evolution must work at near-neutral pH in a salty, protein-rich medium, must not leach metals at concentrations that poison the cells, and must not foul as biofilm forms on it. Moon and colleagues' Cu/NiMo composite catalyst is an example of that specification being treated as a materials problem rather than a biological one, and Lee and colleagues' integrated system making microbial protein from CO_2 with recycled nitrogen is an example of what the architecture is for.

Waste carbon, and the table you would defend

The other route to cheap carbon is waste. Firmansyah and colleagues' approach is a hybrid: oxidise polyethylene waste chemically to short-chain dicarboxylic acids, then feed those to an organism that converts them to a biodegradable polymer. The division of labour is the one from chapter 53 in another form — chemistry breaks the inert polymer, biology does the selective synthesis — and the engineering questions are what the oxidation actually produces, in what distribution and with what contaminants, and whether the organism grows on that mixture rather than on the single acid the abstract names.

Feedstock	Pretreatment	Inhibitors	Achievable yield	Cost driver	Footprint
Glucose	None	None	0.478 g/g ceiling for PHB; 0.23–0.30 observed	Sugar price	Agriculture: fertiliser, land use
Lignocellulosic hydrolysate	Acid or steam explosion, enzymatic saccharification	Furfural, HMF, acetate, phenolics	Similar ceiling; lower in practice, mixed sugars consumed sequentially (part 1 chapter 9)	Enzymes and pretreatment capital	Lower than glucose if residues are genuinely waste
Plastic-waste-derived acids	Chemical oxidation	Oxidation by-products, metals from catalysts, unreacted polymer	Depends on the acid distribution; must be measured	Oxidation step and feed consistency	Credits for diverted waste; depends on allocation
CO_2 + H_2	Electrolysis and CO_2 supply	CO and sulfur in industrial CO_2 streams	18–32 kWh/kg; carbon fully conserved	Electricity price and electrolyser capital	Entirely determined by the grid

Of those cells, you can fill the yields and the CO_2 energy figure from your own calculations — they are in this chapter and chapter 56 — and must look up or measure the pretreatment costs, the inhibitor concentrations in a specific hydrolysate, and the acid distribution from a specific oxidation. Knowing which is which is the point of the exercise.

Constraint sheet — CO_2, H_2 and waste carbon

Binding metric	Electricity price, through hydrogen. 18–32 kWh/kg of PHB depending on the organism's ATP yield per H_2.
Break-even	About \$0.04–0.06/kWh against the sugar route's feedstock cost — before electrolyser and CO_2 capital.
Physical ceiling	Gas–liquid transfer: ~1,000 h^{-1} k_La for H_2 and ~1,900 h^{-1} for O_2 at 2 g/L/h, against a few hundred in a stirred tank. Pressurised loop reactors, not fermenters.
Safety	Oxygen held below the limiting oxygen concentration (~5% for H_2 mixtures), which is itself the transfer bottleneck. Formal review before work starts.
Host requirements	Knallgas metabolism with respiratory energy conservation (*C. necator*) rather than an acetogen, if the product needs ATP; tolerance of the gas regime.
Hybrid routes	Chemistry for the inert step, biology for the selective one. Judge them on feed consistency and contaminants, not on the headline conversion.
Where it fails	On the grid price and on gas transfer, not on the pathway — which is why the research frontier is electrodes and reactors rather than enzymes.

Three things people get wrong

- ‘CO_2 is the feedstock.’ Hydrogen is, and therefore electricity. CO_2 is nearly free and is not what the process is buying.
- ‘The stoichiometric floor is the energy demand.’ It counts reducing equivalents only; the ATP for fixation nearly doubles it, and the factor depends on an assumption.
- ‘Hydrogen transfer is the problem.’ Oxygen is, because safety forces its partial pressure down and with it the driving force.

Self-check — answer on paper first

1. Write the CO_2-to-3HB floor and explain why the pathway tally gives a larger hydrogen demand.

$4\ CO_2 + 9\ H_2 \rightarrow C_4H_6O_2 + 6\ H_2O$, which balances in C, H and O. It counts only reducing equivalents, and the pathway tally agrees on those — 13 NADPH consumed minus 4 NADH generated is 9 — but also shows 16 net ATP required, mostly for the Calvin cycle, which the balanced equation does not represent. Supplying that ATP means respiring additional hydrogen with oxygen, adding 5 to 16 more H_2 per unit depending on the ATP yield per hydrogen, so the real demand is 14 to 25 H_2 against the floor's 9.

2. At what electricity price does a CO_2 route match the feedstock cost of a sugar route running at 0.30 g/g, and what does the comparison omit?

The sugar route's feedstock cost is \$0.40/0.30 = \$1.33 per kilogram of product. At 21.7 kWh/kg — the 2 ATP per H_2 case — matching that requires electricity at \$1.33/21.7 ≈ \$0.06/kWh. The comparison omits electrolyser capital, CO_2 capture and compression, the gas handling and recycle system, the safety engineering, and the fact that the CO_2 route's volumetric productivity is likely lower because of gas transfer, which raises its capital charge per kilogram in exactly the way part 4 chapter 49 describes.

3. Why does holding the oxygen fraction at 5% make oxygen rather than hydrogen the transfer bottleneck?

Because the transfer rate is $k_La \cdot C^*$ and C^* is proportional to partial pressure: dropping oxygen from 21% to 5% of the gas phase cuts its saturation concentration from about 0.21 to 0.05 mM, a factor of four, while hydrogen can be supplied at a high partial pressure. So although the oxygen demand is roughly a quarter of the hydrogen demand in moles, the required k_La for oxygen is about twice as large. The safety constraint and the transfer constraint are the same constraint seen twice.

What to open a book for. For the engineering, Doran's mass-transfer chapters and part 3 chapter 48 of this module cover the transport. For the rest, four papers from the lab this week is built around: Firmansyah et al., Chem. Eng. J. (2025) on upgrading polyethylene waste; Lee et al., Water Res. (2025) on microbial protein from CO_2 in an integrated bioelectrochemical system; Moon et al., Adv. Sci. (2024) on a biocompatible cathode catalyst; and a *C. necator* autotrophic-growth review. Read them with the constraint sheet above beside you and one question each.

Closing notes to part 5

The five classes, side by side

Class	Binds on	Ceiling	Titer needed	Host property that decides it
Pharmaceuticals and natural products	Whether the pathway works at all; purity	0.29 g/g and nobody cares	~1 g/L is a process	Eukaryotic membranes and compartments for P450s
Commodity chemicals and amino acids	Yield, then titer	0.60–0.66 g/g, set by redox	50–150 g/L	Native export, tolerance, cheap medium, regulatory history
Fuels	Yield, and it is still not enough	0.41–0.51 g/g	Irrelevant — the margin is the problem	Anaerobic operation and redox balance
PHA and intracellular polymers	Content, rate and yield together	0.478 g/g, with an oxygen duty	Content >70% at >1 g/L/h	Accumulation physiology under nutrient limitation
CO_2 and waste carbon	Electricity price and gas transfer	18–32 kWh/kg	Set by k_La, not by biology	Knallgas metabolism with respiratory ATP

The pattern worth taking from the table: in four of the five classes, the property that decides the outcome is not in the pathway. It is export, tolerance, accumulation physiology, membrane biology or gas transfer — which is exactly why the host chapters come next and why they are worth writing as a comparison rather than as five species descriptions.

On the numbers in this part

Every yield ceiling here was computed from a written-out balance: the terpenoid cofactor tally per C15 unit, the propanediol redox balance with its oxidation fraction, the PHB net equation with its three-cofactor surplus, and the CO_2-to-3HB floor with the Calvin-cycle ATP tally that reproduces it. The fed-batch table in chapter 56 is constructed for this module rather than taken from a paper, so that every derived quantity could be checked; the method for extracting residual biomass, incremental yield and a harvest criterion from it is what transfers, not the numbers. Reported titers — 25 g/L artemisinic acid, micrograms per litre for opioids, 22 g/L isobutanol, 18 g/L and above 100 g/L for butanediol, above 100 g/L for lysine — are as reported in the sources the tracker names, and are marked as reported wherever they appear in a figure.

The electrolysis energy, gas solubilities, limiting oxygen concentration and ATP yield per hydrogen in chapter 57 are stated assumptions, and the chapter's main result is the size of the range they produce rather than any single value inside it.

What is still to come

Part 6 is the host block: weeks 58 to 61 — the yeasts *S. cerevisiae*, *Y. lipolytica* and *K. phaffii*; the Gram-positives *C. glutamicum* and *B. subtilis*; the phototrophs, cyanobacteria and microalgae; and then the panorama with cell-free systems and the 2025–2026 trends. It will be written against the five constraint sheets in this part, with a single comparison table running through it: genetic tools and editing efficiency, growth rate and medium cost, native precursor and cofactor supply, secretion and post-translational capability, compartments and the transport tax they levy, tolerance, native export, and regulatory history.

Two of those weeks also carry a piece of arithmetic that is this module's own transport calculation in a new currency. Week 58's compartmentalisation question — whether putting a pathway in the mitochondrion or peroxisome pays — is part 3 chapter 48's dead-core problem with membranes instead of gel: a local concentration advantage against a transport cost on every substrate and cofactor that crosses. And week 60's photobioreactor is the same calculation again with photons, where light penetrates a dense culture a few millimetres and the dark volume is a dead core by another name. Both will be computed rather than described.